AZ-305 · D3 — Business Continuity

SURVIVAL · recover data → Backup · keep running → DR · smaller RPO/RTO costs more · ASR = active-passive DRaaS

Backup vs DR basics

Recover DATA (PITR, ransomware, accidental delete) → **Backup**

Keep SERVICE running during region/zone failure → **DR / HA**

RPO = data you can lose (replication gap). **RTO** = downtime you can tolerate. Smaller = pricier.

Vault selection

VMs, on-prem (MARS/MABS), Azure Files, SQL/SAP in VM → **Recovery Services Vault**

Blobs, Disks, PostgreSQL Flex, AKS → **Backup Vault**

Both managed in **Backup Center**. Vault soft delete: **14 d default, 180 d max**; Enhanced soft delete on by default (can't disable).

Azure Migrate: tool per goal

Inventory + dependency map first → **Discovery & assessment**

Rehost VMs as-is (any source) → **Server Migration**

Vendor needs native VMware → **Azure VMware Solution**

Modernize app to PaaS → **App Service Migration Assistant**

Branch files resilience

Multi-branch shared files + fast local access → **Azure Files + File Sync**

Single branch fails → **Cloud share / other caches keep working**

Cloud endpoint is source of truth; each branch server caches hot files locally.

DR multi-region pattern (cheap→costly)

RTO hrs-days · no standing compute · cheapest → **Backup & Restore**

RTO hrs · data replicated · compute at failover → **Pilot Light**

RTO ~5-15 min · secondary at min, autoscales → **Warm Standby**

RTO secs · RPO ~0 · both regions serve → **Active-Active**

"Must serve from both regions" / "RTO <60s" → **active-active**. "15 min OK, cheaper" → warm standby.

Azure Site Recovery scope

Azure VMs between regions → ✅

On-prem VMware/Hyper-V/physical → Azure → ✅

SQL Managed Instance cross-region → ❌ **Failover Groups**

PaaS (App Service, SQL DB, Cosmos) → ❌ **built-in geo**

ASR is **active-passive by design**. "DRaaS in Azure" = ASR, always.

Recovery: 3 protection layers

PREVENT deletion → **Resource Locks**

UNDO recent deletion → **Soft delete / Versioning / PITR**

RESTORE from backup → **RSV / Backup Vault**

Soft delete ≠ versioning: soft delete recovers after **delete**; versioning after **overwrite**. Real protection = enable both. Blob PITR needs versioning + change feed + soft delete.

Recovery windows per workload (exam loves these numbers)

Deleted thing	Mechanism	Default	Max
Blob / version	Blob soft delete	7 d	**365 d**
Blob container	Container soft delete	7 d	365 d
Overwritten blob	Blob versioning	until deleted	
Whole storage acct (block blobs)	PITR (needs versioning+change feed+soft delete)	—	365 d
Azure Files content	Share/file soft delete + snapshots	7 d	365 d
Managed disk	Disk soft delete (sub setting)	7 d	14 d
Key Vault / secret / key	Soft delete + purge protection	90 d	90 d fixed
Entra user / group	Recycle bin	30 d	30 d fixed
Azure SQL DB / MI	PITR (+ LTR)	7 d	35 d / LTR 10 yrs
Cosmos DB	Continuous backup (PITR)	—	30 d
Backup item in vault	Vault soft delete	14 d	180 d

Key Vault purge protection: once on, can't be turned off, items can't be purged before retention (PCI/FedRAMP). **Locks don't protect data inside** (CanNotDelete on account ≠ blocks blob delete). **Immutable blob policy** is container-level data protection: time-based retention = N days/years; legal hold = until cleared; not RBAC.

SQL HA (in-region)

Need	Use
Auto-failover in region (PaaS)	**Business Critical / Premium**
Shared-storage cluster on VMs	**FCI on SQL on VM**
Survive a zone, no app change	**Zone-redundant (GP/BC)**
Read scale-out, same region	**BC free read replica**

SQL geo-DR (cross-region)

≤4 read secondaries, per-DB, manual → **Active geo-replication**

Stable listener + auto-failover → **Failover group**

Managed Instance geo-DR → **Failover group (only)**

Both **async** → RPO>0. Active geo-rep **not on MI**.